CUSTOMER CHURN EXPLORATORY DATA ANALYSIS

Internship Project | EDA Report

DATASET OVERVIEW

Total customers: 7,043
Total columns: 21
Missing values: 0
Duplicate records: 0

CHURN SUMMARY

Churned customers: 1,869 (26.54%)
Customers retained: 5,174 (73.46%)

AVERAGE CUSTOMER STATISTICS

Average tenure: 32.37 months
Median tenure: 29 months
Average monthly charge: 64.76

KEY FINDINGS

1. Month-to-month customers have a 42.71% churn rate.
2. Fiber optic customers have a 41.89% churn rate.
3. Electronic check users have a 45.29% churn rate.
4. Customers without tech support have a 41.64% churn rate.
5. Customers without online security have a 41.77% churn rate.
6. Customers with 0-12 months tenure have a 47.44% churn rate.
7. Customers with 49-72 months tenure have a 9.51% churn rate.

BUSINESS RECOMMENDATIONS

- Improve onboarding for new customers.
- Consider retention offers for month-to-month customers.
- Investigate customer satisfaction among fiber optic users.
- Review the payment experience for electronic check users.
- Promote technical support and online security services.

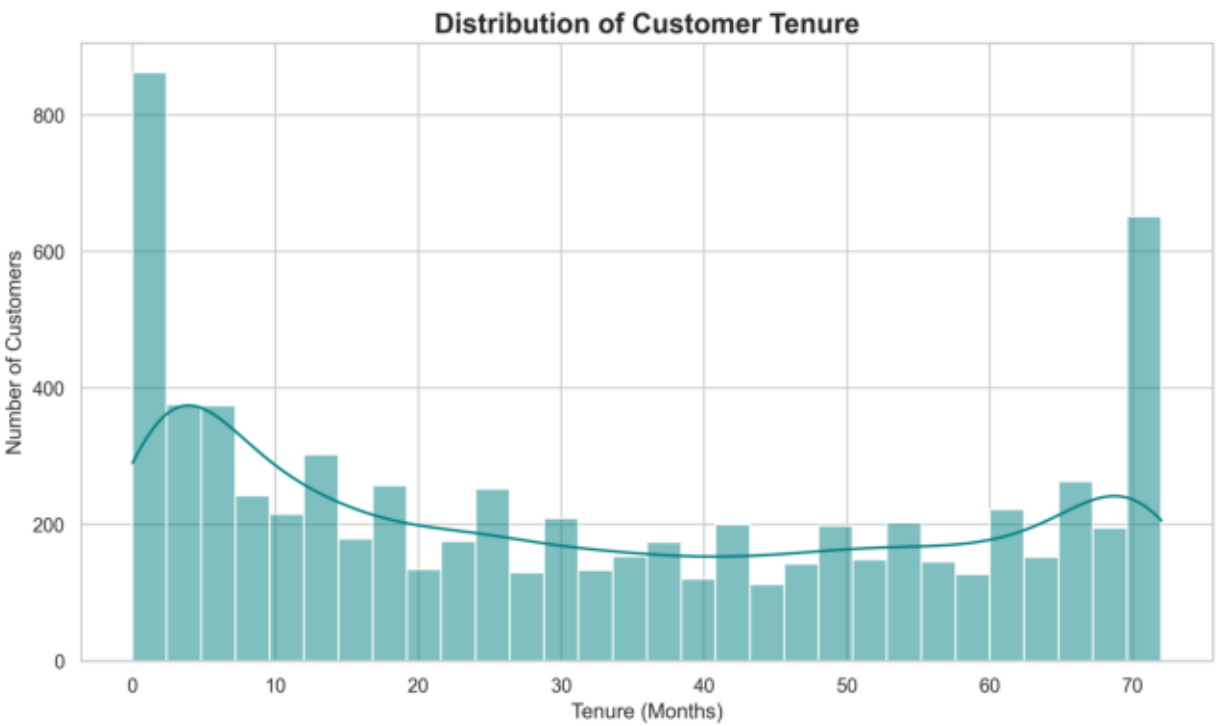
CONCLUSION

The analysis identifies customer groups with higher churn rates.
These patterns can help the business investigate customer retention and improve its services.

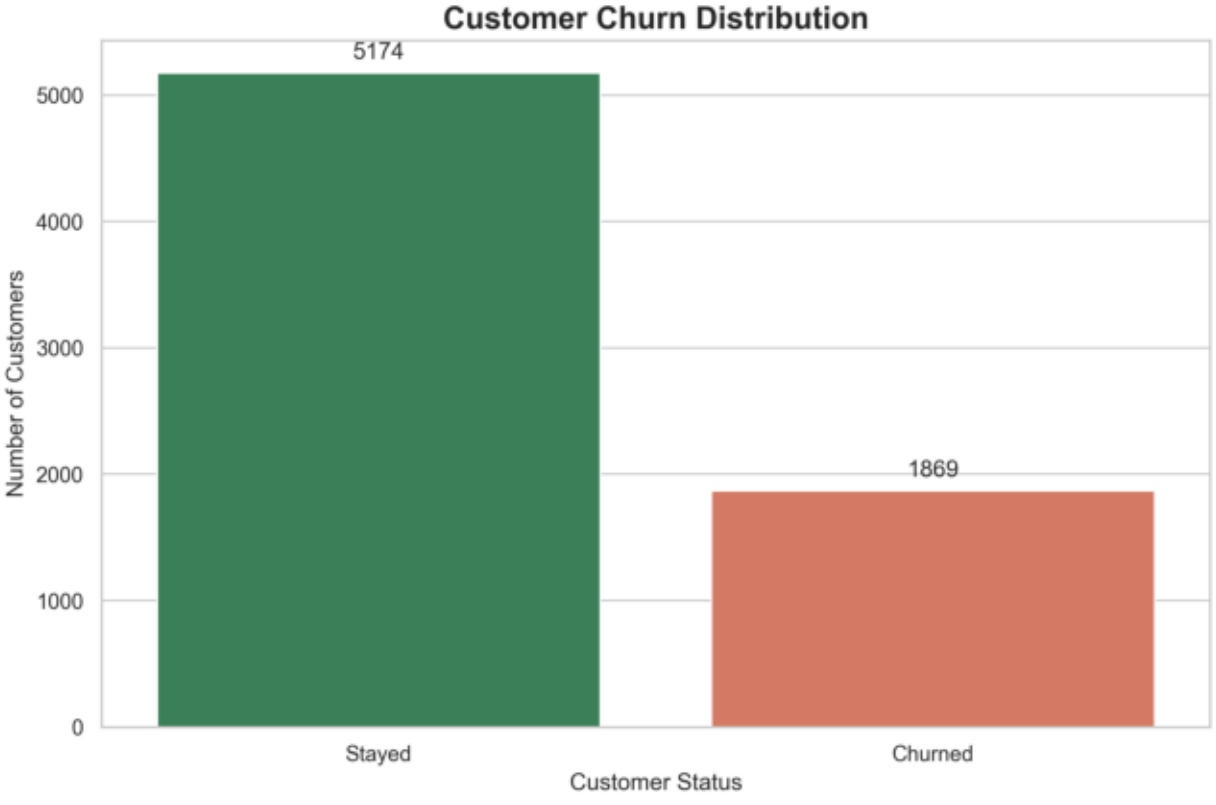
IMPORTANT NOTE

These findings show associations, not proof of causation.
The dataset is the IBM Telco Customer Churn dataset and represents telecommunications customers, not actual boutique transactions.

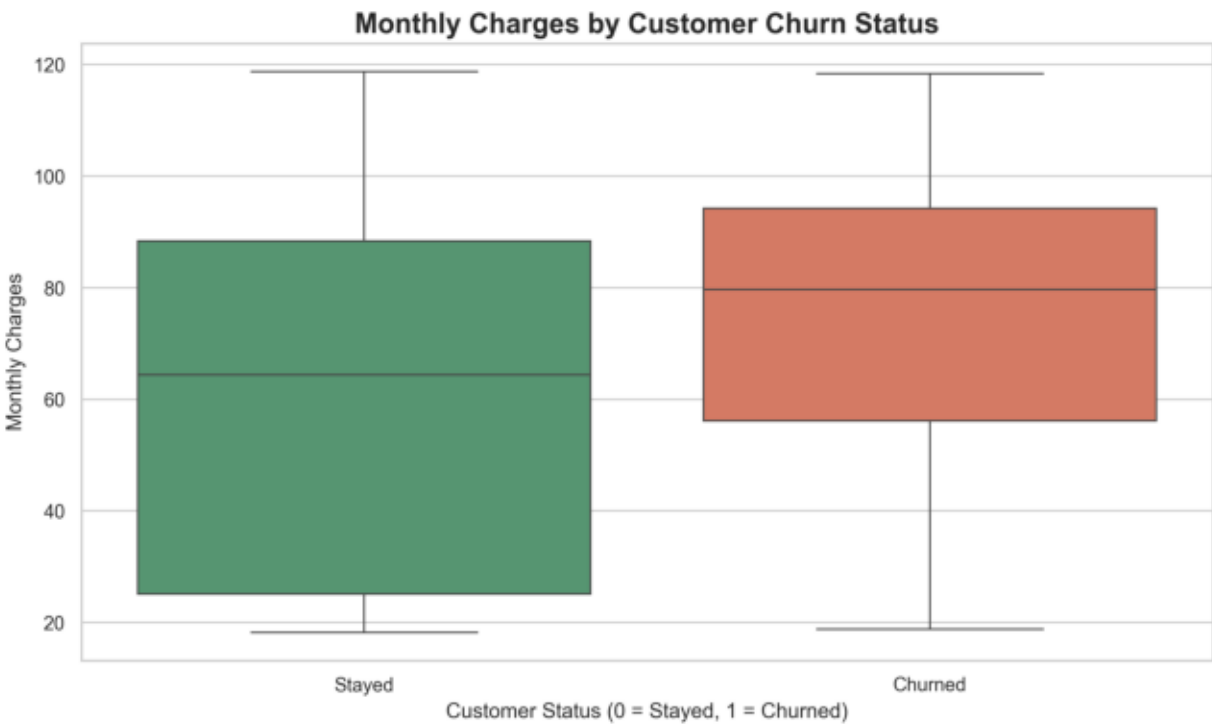
01 Customer Tenure Histogram



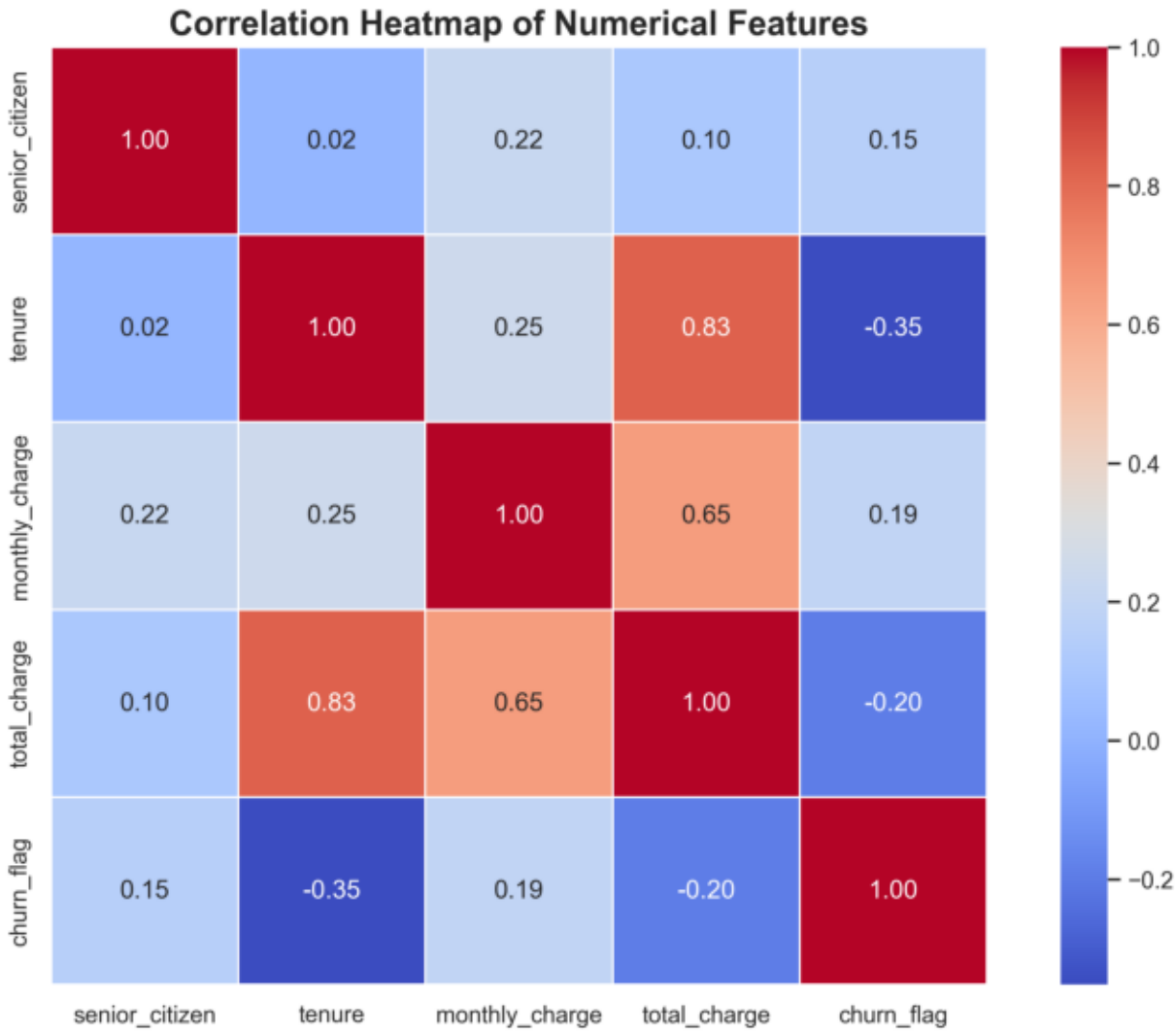
02 Churn Distribution



03 Monthly Charges Boxplot



04 Correlation Heatmap



05 Tenure Vs Total Charges

